Capstone Project Synopsis

AUTOSAR HLD Document Analysis Assistant

Field	Details
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Programme	CSE

1. Problem Statement

AUTOSAR High-Level Design (HLD) documents describe the software components, ports, interfaces, signals and timing of automotive ECUs. They are long and change between revisions, and reviewers check them by hand. Inconsistencies such as a data-type mismatch between connected ports, an undefined interface or a timing requirement that contradicts a runnable period are easy to miss and costly when found late.

2. Objective

To build an assistant that reads HLD documents, extracts the architecture, flags inconsistencies for human review, and answers questions with citations to the exact page and section, while leaving every design decision to the engineer.

3. Intended Users

- Software architects and design reviewers
- Integration and test engineers
- CI maintainers who want an automated design gate

4. Proposed Approach

- Ingest PDF, DOCX, TXT and ARXML files, keeping page and section numbers.
- Extract components, ports, interfaces, signals, runnables and connections with rules, so each item traces to source text.
- Run 15 consistency checks and produce findings for reviewer decision.
- Retrieval-augmented question answering: TF-IDF retrieval, citations, refusal when evidence is weak, and an optional LLM limited to retrieved excerpts.
- Revision comparison, requirement tracing, change impact, test-case skeletons and an exportable review report.

5. Input Data / Knowledge Base

User-supplied HLD documents. Sample set: brake_system_hld_v1.txt (seeded with defects), brake_system_hld_v2.txt (revision) and brake_system.arxml.

6. Technology Stack

Python, scikit-learn, pypdf, python-docx, SQLite, Streamlit (UI), FastAPI (REST API), pytest. Optional OpenAI-compatible LLM.

7. Evaluation Plan

- Automated tests (27) and seeded-defect detection on v1.
- Q&A probe covering answerable and out-of-scope questions.
- v1 to v2 revision comparison; reviewer-based finding precision.

8. Responsible AI

Grounded and cited outputs, local processing, human approval gate for errors, audit logging, and clearly stated limitations.

9. Expected Outcome

A working tool (UI, API and CLI), technical report, demo video and evaluation results.